Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	04 april 2026
Team ID	SWTID-2026-2136
Project Name	Automating Employee Data
Maximum Marks	4 Marks

A	B	C	D	E	F
S.No	Component	Description	Technology		
1	User Interface	Web-based interface for data management and Incident tracking.	ServiceNow Service Portal / Native UI (AngularJS)		
2	Application Logic-1	Logic for importing and validating external data files.	ServiceNow Import Set Engine / JavaScript		
3	Application Logic-2	Logic for mapping staging data to target system tables.	ServiceNow Transformation Engine		
4	Application Logic-3	System-wide rules for field overrides and form behavior.	ServiceNow Dictionary Overrides / UI Policies		
5	Database	Relational storage for employee and incident records.	MariaDB (Internal ServiceNow Database)		
6	Cloud Database	High-availability cloud-based data storage.	ServiceNow Advanced High Availability (AHA)		
7	File Storage	Storage for uploaded CSV/Excel/XML source files.	ServiceNow Staging Table Storage		
8	External API	Integration point for external data sources.	ServiceNow REST/SOAP API		
9	Infrastructure	Global cloud infrastructure for application deployment.	ServiceNow SaaS (Cloud-native)		

A	B	C	D	E
S.No	Characteristics	Description	Technology	
1	Open-Source Frameworks	Client-side frameworks used for the UI and scripting.	AngularJS, jQuery	
2	Security Implementations	Access controls and data protection mechanisms.	ACLs (Access Control Lists), Role-Based Access	
3	Scalable Architecture	Multi-tier architecture that scales with data volume.	ServiceNow Multi-Instance Architecture	
4	Availability	Redundant systems to ensure constant uptime.	Load Balancers / Data Center Failover	
5	Performance	Efficient processing of data imports and form loads.	Concurrent Import Processing / Indexing	